

Database Design Document

MULTI-TENANT BUSINESS MANAGEMENT SYSTEM SCHEMA SPECIFICATION

1. Design Overview & Architectural Principles

This document details the database architecture designed to satisfy the requirements of the multi-tenant Business Management System. The structural design maps directly to the system's Logical and Physical Entity-Relationship Diagrams (ERDs) and enforces data isolation, operational auditing, and role-based access control.

- **Multi-Tenancy Isolation:** Every core child entity references the businesses master table directly via a foreign key constraint (`business_id`). This guarantees absolute logical isolation of data within a unified shared database architecture.
- **Inventory Integrity:** Stock quantities within the `products` table are programmatically tracked through transaction logs within the `inventory_transactions` audit engine, reflecting historical mutations from sales, returns, and purchase orders.
- **Granular Security Architecture:** Employs a robust Role-Based Access Control (RBAC) mapping structure separating user instances from custom authorization tables (`roles`, `permissions`, and the associative resolver `role_permissions`).

2. Data Dictionary & Table Schemas

2.1 Module: Tenant Core Configuration

Table: `businesses`

Stores master settings, metadata, and branding footprints for individual corporate tenants.

COLUMN NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION / REFERENCES
<code>business_id</code>	UUID / Serial	PK	Unique internal identifier for the tenant enterprise.
<code>business_name</code>	VARCHAR(255)	NOT NULL	Legal trading name of the business entity.
<code>phone</code>	VARCHAR(50)	-	Primary telephone contact link.
<code>email</code>	VARCHAR(255)	-	Corporate billing/contact email channel.
<code>address</code>	TEXT	-	Physical headquarters geolocation or post text.
<code>logo</code>	VARCHAR(500)	-	URL string or storage file path pointing to brand media asset.
<code>tax_pin</code>	VARCHAR(100)	-	Government fiscal identifier for localized legal taxation.

COLUMN NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION / REFERENCES
currency	VARCHAR(10)	DEFAULT 'KES'	Default currency tracking metric for transaction summaries.
receipt_footer	TEXT	-	Customized tail text printed at the base of client receipts.
status	VARCHAR(50)	NOT NULL	Operational state matrix flag (e.g., 'Active', 'Suspended').
created_at	TIMESTAMP	DEFAULT NOW()	Record instantiation timestamp.
updated_at	TIMESTAMP	-	System metadata track reflecting last row updates.

2.2 Module: Identity & Role-Based Access Control (RBAC)

Table: roles

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
role_id	UUID / Serial	PK	Primary role key.
role_name	VARCHAR(100)	NOT NULL, UNIQUE	System access title token (e.g., 'Owner', 'Cashier').
description	TEXT	-	Detailed scope statement describing structural permissions.
created_at	TIMESTAMP	DEFAULT NOW()	Creation tracking record.

Table: permissions

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
permission_id	UUID / Serial	PK	Primary permission key identifier.
permission_name	VARCHAR(150)	NOT NULL, UNIQUE	System action flag string (e.g., 'products:create', 'sales:void').
description	TEXT	-	System descriptive path explaining access gate rules.
created_at	TIMESTAMP	DEFAULT NOW()	System time log tracking creation.

Table: role_permissions

Associative junction table establishing a clean many-to-many baseline for operational roles and distinct capabilities.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
role_permission_id	UUID / Serial	PK	Surrogate mapping key identifier.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
role_id	UUID / Serial	FK	References roles (role_id) on cascade delete bounds.
permission_id	UUID / Serial	FK	References permissions (permission_id) on delete cascades.
created_at	TIMESTAMP	DEFAULT NOW()	Assignment creation log data.

Table: users

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
user_id	UUID / Serial	PK	Primary system tracking key for personnel logs.
business_id	UUID / Serial	FK	Scopes instance data directly to parent entity businesses.
role_id	UUID / Serial	FK	Enforces operational access rules linking to roles.
full_name	VARCHAR(255)	NOT NULL	Complete user identifier text.
email	VARCHAR(255)	NOT NULL, UNIQUE	Unique access email token matching auth login targets.
phone	VARCHAR(50)	-	Personnel telephone contact string.
password_hash	VARCHAR(255)	NOT NULL	Secure cryptographically hashed credentials payload text.
status	VARCHAR(50)	NOT NULL	Status check toggling execution loops (e.g., 'Active', 'Disabled').
created_at	TIMESTAMP	DEFAULT NOW()	Initialization timestamp.
updated_at	TIMESTAMP	-	Audit check point capturing parameter alterations.

2.3 Module: Inventory, Categories, & Supply Chain

Table: categories

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
category_id	UUID / Serial	PK	Primary identification key.
business_id	UUID / Serial	FK	Resolves tenant boundary isolation rules back to businesses.
category_name	VARCHAR(150)	NOT NULL	Structured group name mapping tag (e.g., 'Electronics').
description	TEXT	-	Explanatory context data detailing group parameters.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
created_at	TIMESTAMP	DEFAULT NOW()	Instantiation trace.
updated_at	TIMESTAMP	-	Last structural trace logic update tracking metrics.

Table: suppliers

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
supplier_id	UUID / Serial	PK	Primary entity identifier key for suppliers.
business_id	UUID / Serial	FK	Enforces multi-tenant context constraints tracking logs.
supplier_name	VARCHAR(255)	NOT NULL	Commercial organizational identity string.
contact_person	VARCHAR(150)	-	Point-of-contact focal actor tracking identity.
phone	VARCHAR(50)	-	Direct call or system outreach link.
email	VARCHAR(255)	-	Procurement messaging point-of-contact email address.
address	TEXT	-	Corporate mailing index location block.
status	VARCHAR(50)	NOT NULL	Activity mapping vector (e.g., 'Active', 'Inactive').
created_at	TIMESTAMP	DEFAULT NOW()	Creation tracking record.
updated_at	TIMESTAMP	-	Modification metric track execution index point.

Table: products

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
product_id	UUID / Serial	PK	Primary asset item identification index.
business_id	UUID / Serial	FK	References parent businesses(business_id) model.
category_id	UUID / Serial	FK	Categorizes data structure maps via categories links.
supplier_id	UUID / Serial	FK, Nullable	Tracks standard sourcing routes via suppliers index.
product_name	VARCHAR(255)	NOT NULL	Human-readable consumer name asset catalog label.
sku	VARCHAR(100)	-	Stock Keeping Unit track code optimization index.
barcode	VARCHAR(100)	-	Universal Product Code barcode translation target string.
unit_of_measure	VARCHAR(50)	DEFAULT 'pcs'	Tracking unit metric tags (e.g., 'kg', 'box', 'pcs').
cost_price	DECIMAL(12,2)	NOT NULL	Acquisition cost parameter mapping for profit checks.
selling_price	DECIMAL(12,2)	NOT NULL	Standard user consumer catalog list retail pricing point.
stock_quantity	DECIMAL(12,2)	DEFAULT 0.00	Real-time quantitative tracking snapshot metric.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
reorder_level	DECIMAL(12,2)	DEFAULT 0.00	Low-stock alert ceiling trigger marker limit.
status	VARCHAR(50)	NOT NULL	Toggles display logic structures (e.g., 'Active', 'Archived').
created_at	TIMESTAMP	DEFAULT NOW()	Initialization date track logic pointer.
updated_at	TIMESTAMP	-	System metadata track reflecting last row updates.

2.4 Module: Procurement & Supply Chain Flow

Table: purchase_orders

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
purchase_order_id	UUID / Serial	PK	Primary invoice tracking key.
business_id	UUID / Serial	FK	Binds logistics context arrays to businesses.
supplier_id	UUID / Serial	FK	Establishes fulfillment endpoint link to suppliers.
ordered_by	UUID / Serial	FK	Links directly back to purchasing manager inside users.
order_date	TIMESTAMP	NOT NULL	Deployment timeline timestamp signature.
expected_delivery_date	TIMESTAMP	-	Estimated fulfillment log arrival metric window.
total_amount	DECIMAL(12,2)	NOT NULL	Cumulative invoice cost compilation.
status	VARCHAR(50)	NOT NULL	Logistics status (e.g., 'Draft', 'Ordered', 'Received', 'Void').
remarks	TEXT	-	Free text logging vendor exceptions or warehouse comments.
created_at	TIMESTAMP	DEFAULT NOW()	Instantiation date.
updated_at	TIMESTAMP	-	Last modification lifecycle timestamp metric marker.

Table: purchase_order_items

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
purchase_order_item_id	UUID / Serial	PK	Granular line item tracking surrogate index.
purchase_order_id	UUID / Serial	FK	Cascade bounds mapping link back to purchase_orders parent.
product_id	UUID / Serial	FK	Resolves target inventory item linking to products.
quantity	DECIMAL(12,2)	NOT NULL	Total count of resource items ordered from supply source.
unit_cost	DECIMAL(12,2)	NOT NULL	Locked item procurement billing unit price checkpoint.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
subtotal	DECIMAL(12,2)	NOT NULL	Calculated formula array tracking line net cost sums.
status	VARCHAR(50)	-	Fulfillment state parameter checks (e.g., 'Pending', 'Damaged').

2.5 Module: Core Transaction Processing Engine (Point of Sale)

Table: customers

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
customer_id	UUID / Serial	PK	Primary profile tracking key identifier index.
business_id	UUID / Serial	FK	Binds profile instance trace structures to businesses.
full_name	VARCHAR(255)	NOT NULL	Customer descriptive label name string.
phone	VARCHAR(50)	-	Contact and dynamic lookup metric target index field.
email	VARCHAR(255)	-	Receipt distribution target delivery route parameter.
address	TEXT	-	Physical shipping or invoicing geographic description.
created_at	TIMESTAMP	DEFAULT NOW()	Instantiation trace metric marker logic tag.
updated_at	TIMESTAMP	-	Last structural update tracking point sequence map.

Table: sales

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
sale_id	UUID / Serial	PK	Primary ledger entry transaction tracking ID marker.
business_id	UUID / Serial	FK	Enforces core pipeline data isolation scoping to businesses.
cashier_id	UUID / Serial	FK	Traces execution actor directly to operational instance in users.
customer_id	UUID / Serial	FK, Nullable	Links profile in customers; left NULL for walk-in accounts.
sale_date	TIMESTAMP	DEFAULT NOW()	Definitive transactional timestamp marking closing loops.
subtotal	DECIMAL(12,2)	NOT NULL	Gross revenue aggregate pre-deduction valuation total.
discount	DECIMAL(12,2)	DEFAULT 0.00	Total value deduction scalar mapped to absolute gross values.
total_amount	DECIMAL(12,2)	NOT NULL	Net invoice revenue baseline extracted via final formula loops.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
payment_status	VARCHAR(50)	NOT NULL	State vector flag tracking collection (e.g., 'Paid', 'Partially_Paid').
status	VARCHAR(50)	NOT NULL	Operational state check array values (e.g., 'Completed', 'Voided').
created_at	TIMESTAMP	DEFAULT NOW()	Database layer tracking anchor record timestamp.
updated_at	TIMESTAMP	-	Modification metric tracking step value point.

Table: sale_items

Associative split resolving the many-to-many link between the sales transaction engine and the product catalog.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
sale_item_id	UUID / Serial	PK	Granular line item surrogate primary track key tracking index.
sale_id	UUID / Serial	FK	Foreign key cascade link tracing back directly to sales parent.
product_id	UUID / Serial	FK	Establishes direct operational connection back to products.
quantity	DECIMAL(12,2)	NOT NULL	Total count units extracted from available stock channels.
unit_price	DECIMAL(12,2)	NOT NULL	Snapshot tracking actual transaction rate capture values.
discount	DECIMAL(12,2)	DEFAULT 0.00	Line specific deduction tracking scalar parameter check.
subtotal	DECIMAL(12,2)	NOT NULL	Net computed value output baseline calculation: (Qty * Price) - Disc.

2.6 Module: Fiscal Settlement, Receipts, & Material Logs

Table: payments

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
payment_id	UUID / Serial	PK	Primary validation key for transaction settlements.
sale_id	UUID / Serial	FK	Links back directly to targeted invoice parent inside sales.
payment_method	VARCHAR(50)	NOT NULL	Settlement channel trace flag (e.g., 'Cash', 'M-Pesa', 'Card', 'Bank').
amount_paid	DECIMAL(12,2)	NOT NULL	Total numerical monetary cash scale verified by provider.
transaction_reference	VARCHAR(150)	-	External payment system validation reference key (e.g., M-Pesa ID).

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
payment_date	TIMESTAMP	DEFAULT NOW()	Definitive processing timeline confirmation trace log marker.

Table: receipts

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
receipt_id	UUID / Serial	PK	Primary validation log tracker index.
sale_id	UUID / Serial	FK, UNIQUE	Enforces 1:1 transaction integrity mapping constraint to sales.
receipt_number	VARCHAR(100)	NOT NULL, UNIQUE	Formatted, consumer-facing reference invoice ledger alphanumeric.
delivery_method	VARCHAR(50)	NOT NULL	Channel mapping method vector (e.g., 'Print', 'Email', 'WhatsApp').
receipt_date	TIMESTAMP	DEFAULT NOW()	Public physical dispatch confirmation timeline step point.
qr_code	TEXT	-	Dynamic verification payload text string or encoded asset path link.

Table: inventory_transactions

Centrally records every quantitative item variance across the supply network to build an immutable audit history.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
transaction_id	UUID / Serial	PK	Primary immutable ledger index event log tracker key.
business_id	UUID / Serial	FK	Enforces mandatory tenant grouping layer isolation loops.
product_id	UUID / Serial	FK	Maps item mutation back to target inventory asset in products.
user_id	UUID / Serial	FK	Traces executing processing supervisor back to users system.
sale_id	UUID / Serial	FK, Nullable	Optional relationship path linking to invoice in sales if mutation stems from order checkout.
purchase_order_id	UUID / Serial	FK, Nullable	Optional relationship path linking to purchase_orders if variance maps to replenishment intake loops.
transaction_type	VARCHAR(100)	NOT NULL	Action catalog type array (e.g., 'Sale', 'Purchase', 'Return', 'Adjustment').
quantity	DECIMAL(12,2)	NOT NULL	Absolute numerical volume delta shift introduced by tracking step.
quantity_before	DECIMAL(12,2)	NOT NULL	Pre-event baseline balancing point value track check.

COLUMN	TYPE	CONSTRAINTS	DESCRIPTION
quantity_after	DECIMAL(12,2)	NOT NULL	Post-event alignment balancing outcome resolution check trace.
reference_number	VARCHAR(100)	-	Secondary correlation document key reference tracking record string.
remarks	TEXT	-	Context log entry for manual adjustments (e.g., 'Damaged stock').
transaction_date	TIMESTAMP	DEFAULT NOW()	System execution timestamp marking ledger locking sequences.

3. Relational Architecture & Referential Integrity

The system enforces explicit relational rules to ensure data reliability across multi-tenant boundaries. The following baseline configuration principles are hardcoded into the data layer:

- **Cascading Actions:** Multi-tenant context models operate under strict ON DELETE RESTRICT paradigms for key transactional anchors. A tenant business enterprise profile (businesses) cannot be expunged if relational records exist within dependencies like sales or products.
- **Associative Resolvers:** Junction patterns (role_permissions, sale_items, and purchase_order_items) employ strict ON DELETE CASCADE rules targeting their explicit master nodes, preventing orphan pointers from lingering in deep relational loops.
- **Nullability Contexts:** Optional lookup linkages—such as customer_id in the sales ledger—are explicitly set to NULL to natively allow guest/anonymous point-of-sale transactions without breaking tracking boundaries.